

The Definition of the Derivative



Using the limit definition

- 1 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 3x^2 - x$.
 - 2 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+2}$.
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The derivative as a limit at a point

- 3 Use $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ to find $f'(3)$ for $f(x) = x^2 - 2x$.
 - 4 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3$.
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Differentiability and graphical interpretation

- 5 Explain why $f(x) = |x - 2|$ is not differentiable at $x = 2$, computing the one-sided derivatives.
 - 6 State three conditions under which a function fails to be differentiable at a point, with a brief description of each (e.g. corner, cusp, vertical tangent, discontinuity).
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Power, sum, and constant rules

- 7 Differentiate each function using the power rule:
a $f(x) = 7x^5$ **b** $f(x) = \frac{4}{x^2}$ **c** $f(x) = 6\sqrt[3]{x}$ **d** $f(x) = 9$
 - 8 Differentiate $f(x) = 5x^4 - 3x^3 + 2x - 7$.
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Tangent lines from the derivative

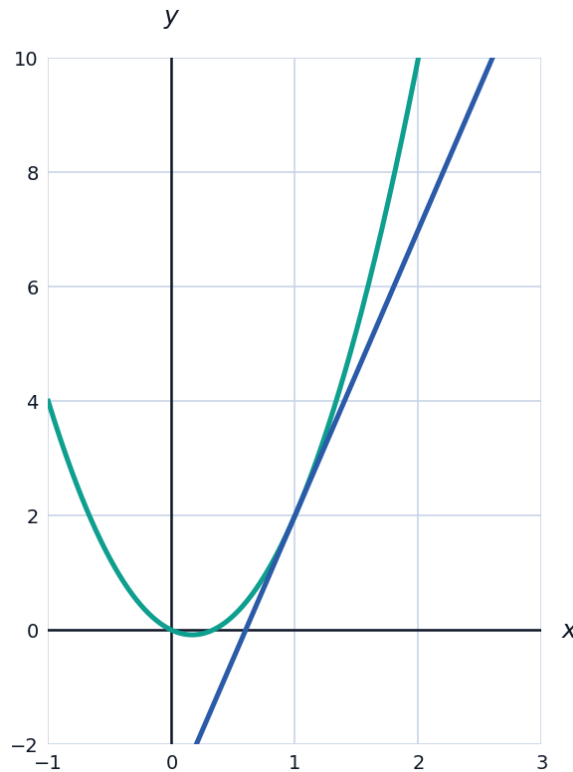
- 9 Using the limit definition result $f'(x) = 6x - 1$ found earlier for $f(x) = 3x^2 - x$, find the equation of the tangent line to f at $x = 1$.
 - 10 Find the equation of the tangent line to $f(x) = x^3$ at $x = 2$, using $f'(2) = 12$ (found earlier).
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Estimating derivatives numerically

- 11 Given $f(1.9) = 3.42$ and $f(2.1) = 3.98$, use a symmetric difference quotient to estimate $f'(2)$.
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Visualizing the derivative as a tangent slope

- 12 The graph shows $f(x) = 3x^2 - x$ (from an earlier problem) with its tangent line at $x = 1$, where $f'(x) = 6x - 1$.



- a Confirm the tangent line's slope matches $f'(1)$.
- b Explain what it means graphically for the line to be "tangent" at this point.
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More practice with the limit definition

- 13 Use the limit definition to find $f'(x)$ for $f(x) = \sqrt{x + 1}$.
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The derivative in applied motion contexts

- 14 A ball's height is $h(t) = -5t^2 + 30t + 1$ (meters, seconds). Use the limit definition of the derivative (or the power rule, since it agrees) to find $h'(t)$, then find the ball's velocity at $t = 2$.

- 15 For the same height function, find the time at which the ball's velocity is 0 (its highest point).
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Comparing average and instantaneous rate of change

- 16 For $f(x) = x^2$, find the average rate of change on $[1, 3]$, then compare it to the instantaneous rate of change $f'(2)$ at the midpoint $x = 2$.